Thin-film Integrated Magnetic Component for Three-level LLC Resonant Converter

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Abstract—**In order to meet the needs of switch power supply with high-power density, low-profile and high-efficiency posed by large scale integration (LSI), this paper presents a novel three-level soft switching DC/DC resonant converter with integrated magnetic technology to reduce its losses, volume and weight, to improve its filter effect and increase its power density. In this paper, equivalent circuit of integrated magnetic components is derived and its working modes are discussed. Thin-film technology is used to design integrated magnetic components, then its design results are given.. At last, its simulation results proved the validity of this design.**

I INTRODUCTION

Electronic products become more and more smaller, thinner, lighter and shorter [1], this trend calls for power supply with high-frequency, high-power density, high-efficiency, low-profile and small volume. Generally, people increase its frequency to reduce the volume, soft-switching technology and using MOSFET as its main switch are the two common ways [2]. But when the input voltage is too high, not only the price of MOSFET is too high but also its performances descend greatly. Then three-level structure is invented to solve this problem. Three-level structure could decrease half the voltage stress of main switch. It is usually used with soft-switching technology to reduce switch loss, but the complexity of circuit increased. At the same time, it has the problem of reverse recovery current on the diode at secondary side. Ordinarily this diode is turned off by force, which brings large current change rate, then loss and withstanding voltage increase greatly.

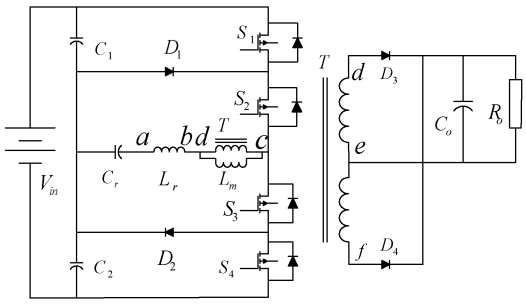

Fig. 1.circuit of three-level LLC resonant converter

Three-level LLC resonant converter show in Fig.1 could solve all the above problems, all its main switches can achieve ZVS at every time, without any ancillary circuit, the diode at secondary side of three-level structure DC/DC converter will work at ZCS at any time. There is no reverse recovery problem, and then voltage peak disappeared.

In this paper, a three-level LLC resonant converter with following specifications will be designed as an example.

Input voltage range V_{in}: 600~800V.
Output voltage range V_{out}: 48V.
Output power range Po: 1.2kW.
Transformer turn ratio: 4:1:1.
Resonant inductor Lr: 14uH.
Resonant capacitor Cr: 0.047uF.
Magnetizing inductance Lm: 60uH.
Switch frequency range: 150~200kHz.

II DERIVATION OF INTEGRATED MAGNETIC COMPONENT'S EQUIVALENT CIRCUIT

Aim of magnetic integration is to reduce the losses, the volume and the mass of magnetic component and to improve its filtering effect [3]. In order to reduce losses and volume of magnetic components, this paper brings up a three-level LLC magnetic integrated converter. The LLT structure in Fig. 1 is similar to duplex winding transformer with consideration of excitation inductor and primary leakage inductor. It is difficult to control leakage inductance while the value of Lr is very important to this resonant converter, an inductor is added to leakage inductor to form a new resonant inductor Lr. Lm is the excitation inductor of the transformer. Therefore, when considering the integration of magnetic components, only an inductor and a transformer are taken into account. In column 2, we cancel the flux generated by inductor and transformer to reduce the volume of integrated magnetics further.

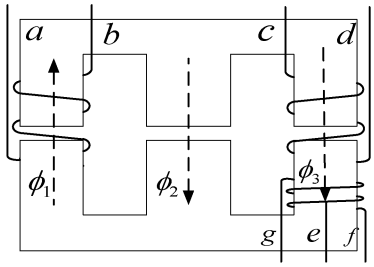

Fig. 2. Integrated magnetic component of three-level LLC resonant converter

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As is shown in Fig. 2, windings of inductor are wrapped on column 1, and its corresponding flux is Φ_1; windings of transformer are wrapped on column 3, and its corresponding flux is Φ_3; the flux of common column 2 is Φ_2.Terminals of integrated magnetics are connected as is shown in Fig.2.

Using the dual relationship in magnetic circuit and faraday's law of electromagnetic induction [4], we can get equivalent circuit with following steps:

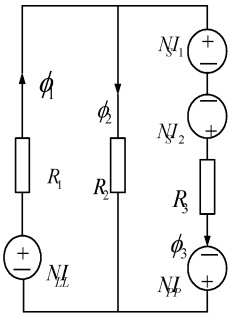

Fig. 3 （a）

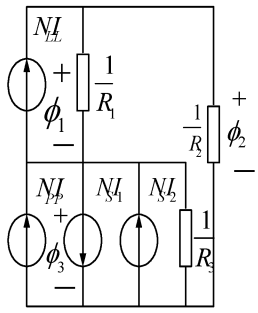

Fig. 3 （b）

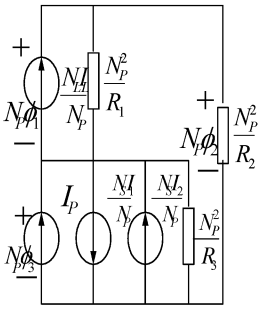

Fig. 3 （c）

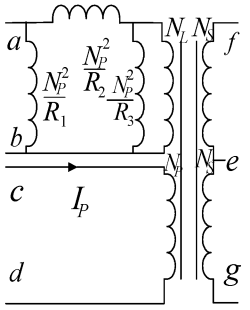

Fig. 3 （d）

Fig. 3（a）equivalent magnetic circuit of integrated magnetic component

Fig. 3（b）dual graph of equivalent magnetic circuit

Fig. 3（c）dual graph of equivalent magnetic circuit after equivalent transform

Fig. 3（d）equivalent circuit of integrated magnetic component

1. According to ohm's law, we can get the equivalent magnetic circuit of integrated magnetic component Fig. 3（a）. N_L, N_P, N_S, N_S, I_L, I_P, I_{o1}, I_{o2} are turn numbers and current of windings ab, cd, ef and eg. R1, R2, R3 are magneto resistance of three column separately.
2. Based on the dual relationship between magnetic circuit and circuit, dual graph of equivalent magnetic circuit could be obtained, as is shown in Fig. 3（b）.
3. Take dimension scale transform to dual graph.
4. In accordance with dual graph of equivalent magnetic circuit after equivalent transform, using faraday's law of electromagnetic induction impedance variation method of transformer, equivalent circuit of integrated magnetic component is obtained.

III ANALYSIS OF THREE-LEVEL LLC RESONANT CONVERTER WITH INTEGRATED MAGNETIC COMPONENT

The schematic of three-level LLC resonant converter is shown in Fig.1, three-level half-bridge structure is formed by four main switches and two capacitors. Inductor Lr, Lm and capacitor Cr form a resonant tank, where Lr is the resonant inductor, Lm is the excitation inductor, this converter uses Variable-frequency method other than PWM to control output voltage. When S_1 and S_4 were powered off, D_1 and D_2 clamp its voltage half the input voltage. In this converter, we consider the output filter capacitor Co big enough, so that its voltage can be seemed as V_o, meanwhile the input capacitors can also be substituted by $V_{in}/2$. The main switches are MOSFET with inverse parallel diodes. Wave form in one cycle is symmetrical, so we can only analysis the half operation modes in former half cycle.

Set: $L_1 = N_P^2/R_1$； $L_2 = N_P^2/R_2$； $L_3 = N_P^2/R_3$；

Operation mode one

Three-level magnetic integration LLC resonant converter's circuit and its equivalent circuit are shown in Fig.4. In this mode, S_1 and S_2 are at conducting state, primary current i_{in} flow through it, wave form of current is sinusoidal, diode D_3 at secondary side will turn on. The output voltage will be clamped at V_o, at this time, resonant will happen between Lr and Cr. Current of Lm will increase linearly.

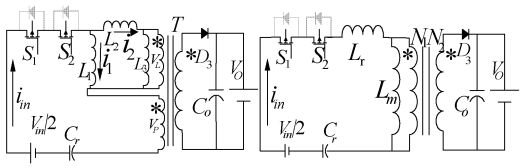

Fig.4 proposed converter's circuit in mode one and its equivalent circuit

For operation mode one, we can get following equations:

$$\frac{1}{2}V_{in} = \frac{1}{C_r}\int i_{in}dt + L_1\frac{di_1}{dt} + V_p \quad (1);$$

$$\frac{1}{2}V_{in} = \frac{1}{C_r}\int i_{in}dt + L_2\frac{di_2}{dt} + V_L + V_P \quad (2);$$

$$V_P = \frac{N_P}{N_S}V_O \quad (3);$$

$$V_L = \frac{N_L}{N_S}V_O \quad (4);$$

From above equations, we can get the relationship of input voltage, input current and output voltage as following:

$$\frac{1}{2}V_{in} = \frac{1}{C_r}\int i_{in}dt + \frac{L_1L_2}{L_1+L_2}\frac{di_{in}}{dt} + \frac{1}{N_S}(N_L + N_P\frac{L_1}{L_1+L_2})V_O \quad (5);$$

From this equation, we can get equivalent circuit during this mode as in Fig.4, and we also can get following equation:

$$\frac{1}{2}V_{in} = \frac{1}{C_r}\int i_{in}dt + L_r\frac{di_{in}}{dt} + \frac{N_1}{N_2}V_O \quad (6);$$

In the equivalent circuit, Lr and N1 are as following:

$$L_r = \frac{L_1 L_2}{L_1 + L_2} \quad (7);$$

$$N_1 = N_L + N_P\frac{L_1}{L_1 + L_2} \quad (8);$$

Operation mode two

With the increase of Lm's current of and decrease of Lr's current, at one time, these two values will equal, then the current of secondary side becomes zero, D_3 will turn off automatically. So there is no reverse recovery process and the voltage on transformer no longer be clamped by output voltage. This resonant happens among Lr, Lm and Cr.

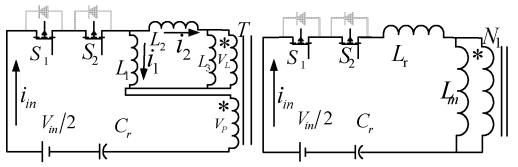

Fig.5 proposed converter's circuit in mode two and its equivalent circuit

According to operation mode two, we can get: :

$$\frac{1}{2}V_{in} = \frac{1}{C_r}\int i_{in}dt + L_1\frac{di_1}{dt} + V_p \quad (9);$$

$$\frac{1}{2}V_{in} = \frac{1}{C_r}\int i_{in}dt + (L_2 + L_3)\frac{di_2}{dt} + V_P \quad (10);$$

$$V_P = V_L N_P / N_L \quad (11);$$

From above equation, we can get relationship of input voltage and input current as following:

$$\frac{1}{2}V_{in} = \frac{1}{C_r}\int i_{in}dt + \frac{L_1}{N_P}\left(\frac{N_P L_1 + N_P L_2 + N_2 L_3}{L_1 + L_2 + L_3}\right)\frac{di_{in}}{dt} \quad (12);$$

From this equation, we can get equivalent circuit during operation mode two as in Fig.5:

$$\frac{1}{2}V_{in} = \frac{1}{C_r}\int i_{in}dt + (L_r + L_m)\frac{di_{in}}{dt} \quad (13);$$

In above circuit, Lm is as following:

$$L_m = \frac{L_1}{N_P}\left(\frac{N_P L_1 + N_P L_2 + N_2 L_3}{L_1 + L_2 + L_3}\right) - L_r \quad (14);$$

Operation mode three

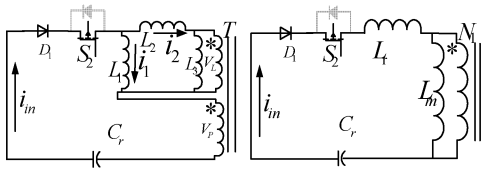

Fig.6 proposed converter's circuit in mode three and its equivalent circuit

S_1 still conduct, S_2 turned off, then the current of Lr and Lm continuous-flow through fly-wheel diode D_1. At this time, the voltage on S_1 will be clamped at half input voltage.

Operation mode four

S_2 also turned off, current on primary side will be transferred to S_3 and S_4, parasitic diode of S_3 and S_4 will be conducted. And then the voltage on S_1 and S_2 is input voltage.

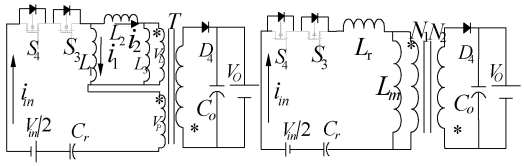

Fig.7 proposed converter's circuit in mode four and its equivalent circuit

From above analysis, the four main switches of three-level LLC resonant converter all achieved ZVS, and each switch's voltage stress is half input voltage. Diodes at secondary side can be turned off while its current is zero, and this could avoid reverse recovery process, and its voltage stress is only half input voltage.

There are two resonant frequencies in this resonant converter, when the resonant tank only contains Lr and Cr, its resonant frequency is

$$f_1 = 1/(2\pi\sqrt{L_r C_r}) \quad (15);$$

when the resonant tank contains Lr, Cr and Lm, its resonant frequency is

$$f_2 = 1/(2\pi\sqrt{(L_r + L_m)C_r}) \quad (16);$$

The maximum exciting current is

$$I_m = nv_0 / 4L_m f_1 \quad (17);$$

Where Lm is exciting inductor, V_o is output voltage, n is transformation ratio. When the switch frequency is lower than f_1, all the above operation modes will happen, but when the switch frequency is equal to f_1, operation mode two will disappear. The voltage of this converter is controlled by variable its switch frequency. Exciting inductor Lm can be seen as a boost inductor, when the resonant become heavier, this boost inductor will make the voltage on resonant capacitor higher, at last, the output voltage increased.

Following is an example of design: N_L=8, Np=4, Ns=2, air gap on each column is 0.56mm.

Ⅳ DESIGN OF THREE-LEVEL MAGNETIC INTEGRATED LLC RESONANT CONVERTER WITH THIN-FILM TECHNOLOGY

Thin-film technology can make magnetic components change from 3D structure to 2D structure, and it will greatly reduce the dimension of magnetic components [5]. In accompany with the development of the soft magnetic material with low loss, high resistance, high saturate permeability, high start permeability, wide operation temperature and high operation frequency, magnetic components

with high operation frequency come true, which provide a new way to achieve "short, small, light and thin" of switching power supply [1].

Option of magnetic material

The core of the transformer and inductor in the proposed three-level LLC resonant converter is bidirection magnetized, that's to say, this core is working under the first condition [1]. Under this condition, working magnetic density of the magnetic core is between B_m and - B_m, in order to reduce the volume and the mass, we should use the material with high saturation magnetic density, since working magnetic density of the magnetic alternant along the hysteresis cycle, core loss is large, especially in high frequency, the influence of skin effect is serious, this will induce large eddy current loss. So we should choose material with narrow hysteresis cycle and high saturate magnetic density in order to reduce exciting current.

Literature [5] poses a good material CoNiZr, which has following merits: high electrical resistivity, high saturation flux density, small hysteresis cycle, low self-losses. It is a good thin-film magnetic material. Based on above advances, this design uses CoNiZr as our magnetic core material.

Design of thin-film inductor

Different design of wrap shape will lead different inductance, different chip area and will determine the complexity of fabrication process. Spiral type inductor, meander type inductor and cloth type structure inductor are the main types of inductor [6].

Compare the pros and corns of all the types, we choose spiral type inductor. Usually, the windings use photoengraving or ion etching method. Considering the cost and complexity, PCB is adopted to fabricate windings. PCB can be used to make not only single-side board but also dual platen, even multi-layer board. Multi-layer board can enhance couple degree, and can get the inductance twice more than single-side due to L=L1+L2+2M. It's an available way to get larger inductance in every unit area. Fig. 8 is the scheme diagram of thin-film windings.

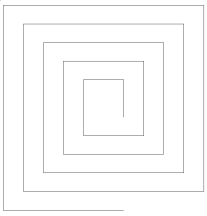

Fig.8 winding structure of thin-film inductor

1) Calculation of copper foil: $w= i_{max}/(hj)$; where h represents the thickness of copper foil, and normally it is less than twice of skin depth; j is the current density; i_{max} is the largest current; w represents the copper foil width.

2) Calculation of isolate dielectric-slab: set the area of isolate dielectric-slab as the area of inductor windings. According to $C=A_L \varepsilon_0 \varepsilon_r / h_r$, the thickness of isolate dielectric-slab h_r will be determined. Where C is the capacitance, ε_0 is the dielectric constant of vacuum, ε_r is the dielectric constant of the dielectric-slab, A_L represents the area of inductor windings.

Set two layers of the thin-film inductor as a basic unit, a_1, b_1, c_1, d_1 are the connecting terminal of upper layer and bottom layer separately. The equivalent circuit of every basic unit is shown in Fig. 9.

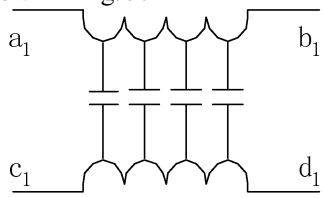

Fig. 9 equivalent circuit of basic unit

The two layers of basic unit can bring capacitor, while the LLC resonant converter needs capacitor Cr, then we want to make use of this capacitor. The distance between two layers is controllable, we can adjust the space between two layers in order to get the capacitor we want.

From above theories, width of copper foil is 3mm, distance between two paralleled copper foils is 0.5mm, its depth is 0.105mm, the distance between two layers is 0.7mm.

Design of thin-film transformer

The main types of thin-film transformer are as follows: thin-film transformer based on common principal, thin-film transformer with coaxial windings, pinhole type thin-film transformer [6]. For the sake of fabrication convenience, we choose thin-film transformer based on common principal. We also use PCB to fabricate its winding structure.

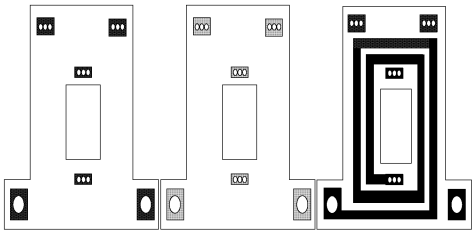

Fig.10 (a) Fig.10 (b) Fig.10 (c)

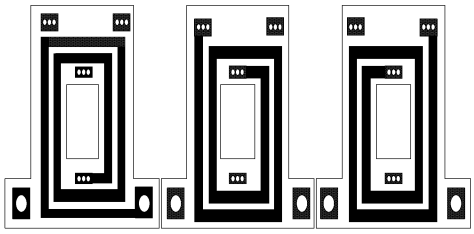

Fig.10 (d) Fig.10 (e) Fig.10 (f)

Fig,10 winding structure of thin-film transformer

Since Np=4, N_L=8, N_2=2, we could get the

transformer ratio 4:1:1. Let winding structure of thin-film transformer the same as is shown in Fig.10, where Fig.10 (a) and Fig.10 (b) are top and bottom layer, Fig.10 (c) and Fig.10 (d) are primary windings of transformer, Fig.10 (e) and Fig.10 (e) are transformer's secondary windings. They are connected by "through-hole".

Design of thin-film integrated magnetics

According to the proposed technology, we could get the appropriate thin-film integrated magnetic component, which satisfied the circuit well, and we use magnetic control sputtering to fabricate the magnetic core.

In order to reduce parasitical capacitor between two units, we stagger the copper foil of each unit just as is shown in Fig.11, and transformer is the same. Since the magnetic core could result large leakage at the side columns, we change the magnetic structure from EI to the shape just like a Chinese character "four" as is shown in Fig. 11.

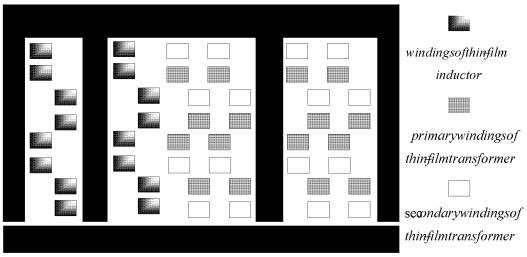

Fig.11 cross-section of thin-film integrated magnetic component

V SIMULATION RESULTS

From above analysis, we use Maxwell SV to do simulation. The parameters of thin-film integrated magnetic component are as follows: the width of copper foil is 3mm, depth is 0.105mm, space between coppers is 0.5mm, air gap is 0.56mm, thickness of isolate dielectric is 0.7mm, height of magnetic core is 10mm. The simulation result is shown as Fig.12.

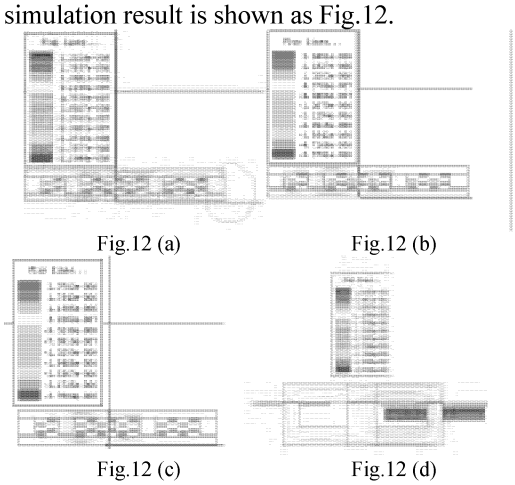

Fig.12 (a) Fig.12 (b)

Fig.12 (c) Fig.12 (d)

Fig.12 simulation results

Fig.12 (a) is the diagram of flux distribution while only current i_{in} flow through the windings of thin-film inductor, Fig.12 (b) is the diagram of flux distribution while only current i_{in} flow through the windings of thin-film transformer, Fig.12 (c) is the diagram of flux distribution while thin-film integrated magnetic component is used in the proposed topology, Fig.12 (d) is the simulation result of traditional integrated magnetic component. Compare these simulation results. We could find that after designed by thin-film technology, the height of integrated magnetic component reduced more, and meet the demand of "thin, short, light, small", further more it adapt the needs of development of SMT technology, and make large scale commercial process possible. In addition, the flux density of thin-film integrated magnetic is large, which made high use of magnetic component. At the same time, it could bring flux saturation, but the connection way of integrated magnetic component's terminals and use of a new type soft magnetic material CoNbZr with high saturation flux density alleviate this contradiction greatly.

VI CONCLUSIONS

This paper designs a thin-film integrated magnetic component for three-level LLC resonant converter, derives the equivalent circuit of integrated magnetic component, discusses its operation modes, lists the design results, and then gives the thin-film integrated magnetic component using thin-film technology, at last, simulation results are given. This design shows a great superiority, makes a highly use of magnetic core and its thickness is only 10mm, which accommodates the demand of SMT technology and achieves "short, small, light and thin" of LLC resonant converter.

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